Pre-eclampsia and eclampsia

Definition

- BP ≥ 140/90 after 20 weeks + proteinuria or organ dysfunction
- Severe: BP ≥ 160/110, platelets < 100, liver/renal impairment, pulmonary oedema, neuro/visual

Eclampsia management

- **Magnesium sulphate 4 g IV over 5–15 min**, then 1 g/h for 24 h (Magpie)
- Recurrent seizures: further 2 g; monitor reflexes, RR, UO, levels (toxicity: calcium gluconate)
- Delivery after stabilisation

BP control

- Target < 150/100 (or 135/85); labetalol, nifedipine, hydralazine

Key point: Magnesium for seizures and urgent BP control for stroke prevention.

Notes: pregnancy p.11

HELLP versus acute fatty liver of pregnancy

Comparison

- HELLP: haemolysis, high liver enzymes, low platelets; with pre-eclampsia
- AFLP: liver failure, hypoglycaemia, coagulopathy, AKI; Swansea criteria

Management

- **Delivery** after stabilisation; correct coagulopathy, glucose
- Magnesium if pre-eclampsia; transplant rarely

HELLP complications

- DIC, abruption, AKI, hepatic haematoma/rupture, pulmonary oedema

Key point: Delivery is the definitive treatment for both.

Notes: pregnancy p.25

Amniotic fluid embolism

Diagnosis

- Sudden hypoxia, hypotension/arrest, coagulopathy during labour/delivery
- Pathophysiology: anaphylactoid response → pulmonary hypertension, RV failure, DIC

Resuscitation

- ALS, left uterine displacement, perimortem CS within 5 min
- RV support (inotropes, iNO), ECMO

Coagulopathy

- Massive transfusion, fibrinogen > 2 g/L, TXA

Key point: AFE: rapid collapse + DIC: treat RV failure and coagulopathy.

Postpartum haemorrhage

Definition and causes

- ≥ 500 mL vaginal, ≥ 1000 mL CS (major > 1500)
- **4 Ts**: tone (70%), trauma, tissue, thrombin

Atonic PPH

- Uterine massage, oxytocin, ergometrine, carboprost, misoprostol
- **TXA 1 g within 3 h** (WOMAN); balloon tamponade, B-Lynch, embolisation, hysterectomy

Fibrinogen

- **< 2 g/L predicts severe PPH**; replace with cryo/fibrinogen concentrate

Key point: Early TXA and fibrinogen replacement save lives.

Notes: pregnancy p.26

Peripartum cardiomyopathy

Definition

- LVEF < 45% in late pregnancy to months postpartum, no other cause
- Risks: multiparity, age, pre-eclampsia, twins, African ancestry

ICU management

- Standard heart failure therapy (avoid ACEI antepartum)
- **Bromocriptine** (blocks prolactin), anticoagulation if EF < 35%
- MCS/ECMO, transplant

Prognosis

- 50% recover; recurrence risk in future pregnancies

Key point: Bromocriptine is specific therapy for PPCM.

Notes: pregnancy p.3

Cardiac arrest in pregnancy

Physiological changes

- ↑ CO, ↑ blood volume, ↓ FRC, ↑ O2 consumption, aortocaval compression after 20 weeks

ALS modifications

- **Manual left uterine displacement**, early airway (smaller ETT), IV access above diaphragm
- **Perimortem caesarean within 5 min** if > 20 weeks and no ROSC

Reversible causes

- Bleeding, AFE, eclampsia, Mg toxicity, cardiac disease, sepsis, anaesthetic complications

Key point: Perimortem CS helps the mother first.

Notes: pregnancy p.13

Maternal sepsis

Causes

- Endometritis, wound infection, UTI, pneumonia, mastitis, septic abortion
- Group A strep most deadly

Recognition challenges

- Physiological changes (tachycardia, low BP, high WBC) mask sepsis; use obstetric scores (MEOWS, omqSOFA)

Management

- Hour-1 bundle; broad antibiotics (pip-taz/carbapenem + clindamycin for toxin)
- Source control, fetal monitoring

Key point: Pregnancy hides sepsis: use obstetric early warning scores.

Respiratory failure in pregnancy

Changes

- ↑ Minute ventilation (respiratory alkalosis), ↓ FRC, $PaCO_2$ 4 kPa normal

Ventilation targets

- $SpO_2 \geq 95\%$, $PaO_2 > 9.3$ kPa, $PaCO_2$ 4–4.7 kPa (fetal)
- Prone possible with care; ECMO if refractory

Delivery timing

- May improve maternal respiratory mechanics; decided with obstetrics
- Influenza: oseltamivir

Key point: Avoid maternal hypoxia: fetal oxygenation depends on it.

Ovarian hyperstimulation syndrome

Pathophysiology

- VEGF-driven capillary leak after hCG (IVF)
- Severe: ascites, haemoconcentration, oliguria, thrombosis

ICU management

- Fluids (albumin), paracentesis, thromboprophylaxis
- Avoid diuretics unless overloaded

Complications

- Thrombosis, AKI, ARDS, pleural effusion, ovarian torsion

Key point: Capillary leak syndrome: careful fluids and thromboprophylaxis.

Hyperemesis gravidarum

Definition

- Severe vomiting, weight loss > 5%, dehydration, electrolyte disturbance
- Complications: Wernicke's, hypoK, hypoNa, Mallory–Weiss

Management

- IV fluids (avoid dextrose before thiamine), **thiamine**, antiemetics
- K, Mg replacement; VTE prophylaxis

Exclude

- Thyrotoxicosis, molar pregnancy, pancreatitis, hepatitis, DKA

Key point: Thiamine before glucose in hyperemesis.